Statistics and Probability Course

Statistics

Statistics Basics

S**tatistics** (Definition)

Types of Statistics

1. Descriptive Statistics
2. Inferential Statistics

Types of Statistical Studies

1. Observational Study
2. Designed Experiment

Sampling Terms

- Population
- Sample
- Sampling
- Sampling Unit
- Sampling Frame
- Census
- Sample Survey
- Parameter
- Statistic
- Representative Sample
- Homogeneity
- Heterogeneity

Sampling Techniques

Probability Sampling

- Simple Random Sampling
- Systematic Random Sampling
- Stratified Sampling
- Cluster Sampling

Non-Probability Sampling (not studied types)

- Convenience Sampling
- Judgement Sampling
- Quota Sampling
- Snowball Sampling

Variables and Data

Variable Terms

Variable

Qualitative Variable

Quantitative Variable

- Discrete Variable
- Continuous Variable

Study Variable Auxiliary Variable

Data & Data Types

Data

- Qualitative Data
- Quantitative Data
- Primary Data
- Secondary Data

Classifications of Data

- Time Series Data
- Cross-Sectional Data
- Panel Data

Presentation of Data

Quantitative Data

- Frequency Distribution Table
- Histogram
- Frequency Curve
- Frequency Polygon

Qualitative Data

- Simple Bar Chart
- Pie Chart
- Multiple Bar Chart (Cross Tabulation)
- Component Bar Chart

Descriptive Measure or Summary Measure

1. Measures of Central Tendency
2. Measures of Variation or Dispersion
3. Measures of Skewness
4. Measures of Kurtosis.

Measure of Central Tendency

1. Arithmetic Mean
2. Geometric Mean
3. Harmonic Mean
4. Median
5. Mode

Measures of Dispersion or Variation

1. Absolute Measures

1. Range
2. Mean Deviation
3. Standard Deviation
4. Variance

2. Relative Measures

1. Coefficient of Range
2. Coefficient of Mean Deviation
3. Coefficient of Variation

Measure of Skewness

- Right-Skewed
- Left-Skewed
- Zero Skewed
- Symmetric

Position Measures

1. Quartile
2. Decile
3. Percentile

Probability